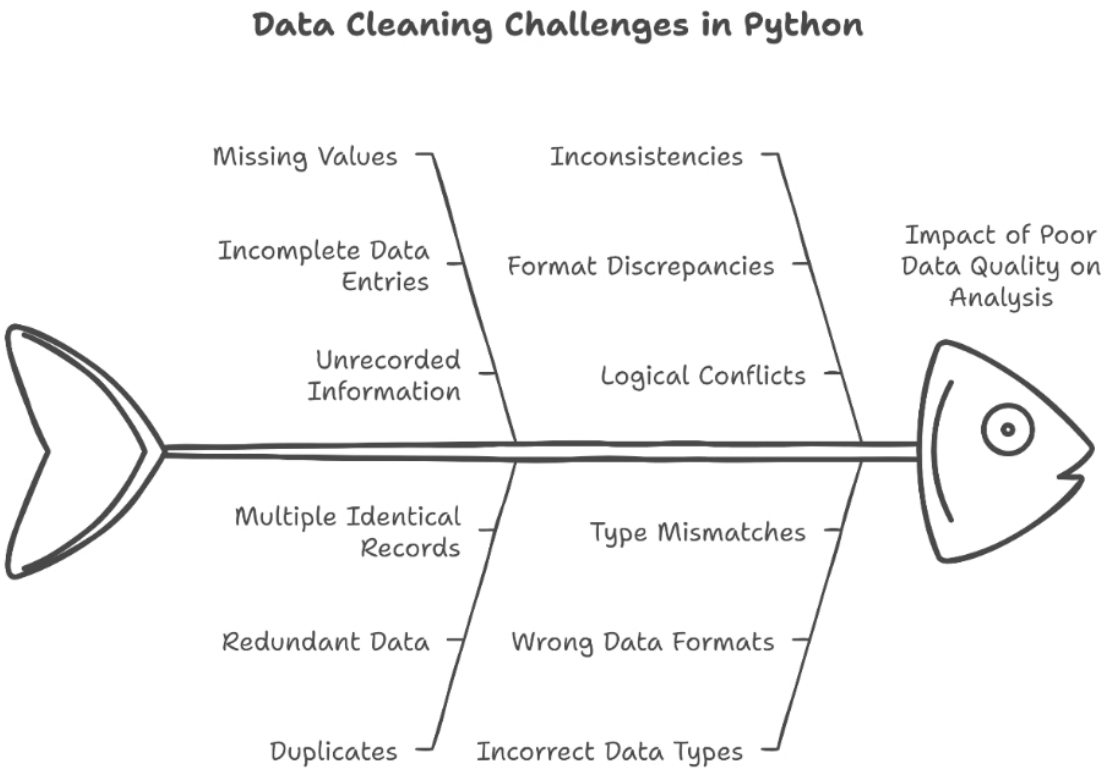
Data Cleaning and Preprocessing in Python

Data cleaning and preprocessing are essential steps in any data analysis or machine learning workflow. Raw data often contains issues such as missing values, duplicates, inconsistencies, and incorrect data types, which can negatively impact the quality of your analysis or model performance. Python, with its versatile libraries like Pandas and Scikit-Learn, provides powerful tools to handle these challenges effectively.



Importance of Data Cleaning

Data cleaning is the process of identifying and correcting (or removing) errors and inconsistencies in data to improve its quality. High-quality data is crucial because it directly impacts the accuracy of your insights and predictions. Even the most sophisticated machine learning models can underperform if fed with unclean data.

Handling Missing Data